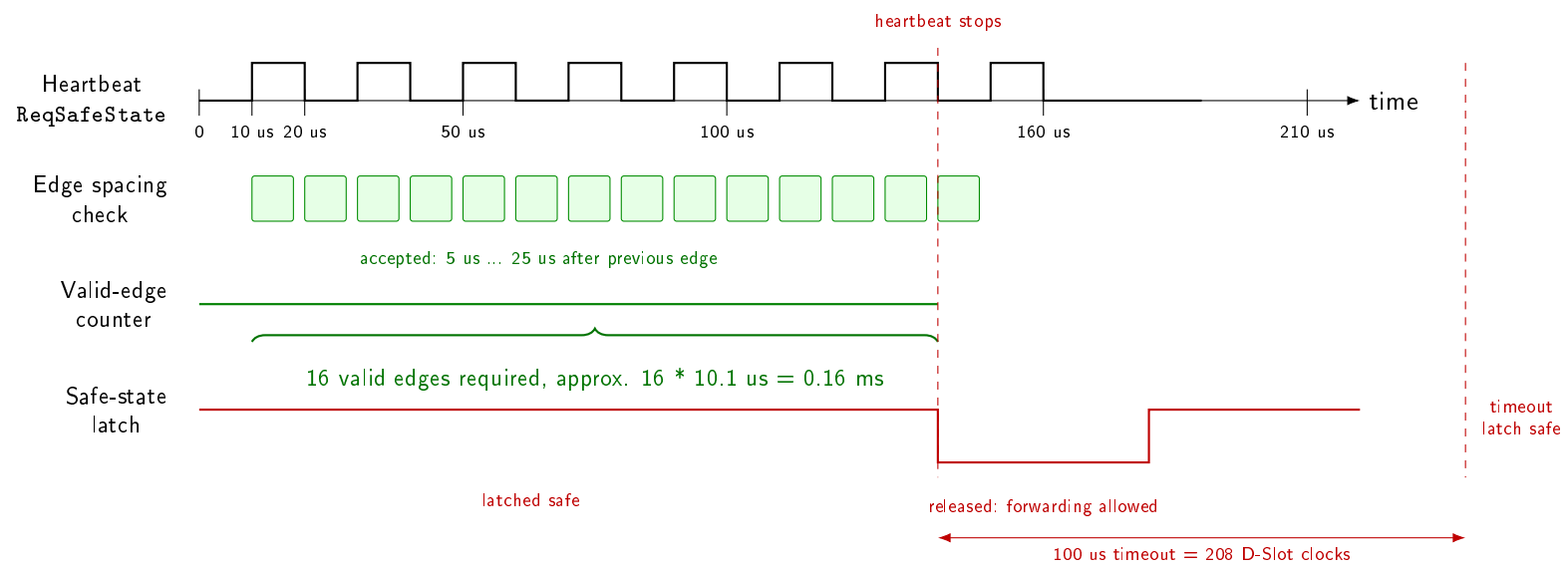


Both CPLDs use the internal 2.08 MHz oscillator. S3C divides this clock to a heartbeat with 21 clock ticks per half period: $21 / 2.08 \text{ MHz} = 10.1 \text{ us}$ per edge, 20.2 us full period, approx. 49.5 kHz. The D-Slot accepts only edges spaced between 5 us and 25 us, so a static line, a slow signal, or a short glitch cannot release the safe-state latch.



D-Slot edge validation
HB_MIN_EDGE_CLKS = CLK/200000 = 10 clocks = 4.8 us
HB_MAX_EDGE_CLKS = CLK/40000 = 52 clocks = 25.0 us
HB_TIMEOUT_CLKS = CLK/10000 = 208 clocks = 100 us
Edge accepted only if previous edge was inside this window.

Fail-safe behavior
Power-up: safe_state_request = 1.
Static 0/1 or missing S3C supply: no valid edges, safe remains latched.
After valid S3C Ready heartbeat: 16 edges release the latch.
Heartbeat loss: within 100 us, latch returns to safe.